



TECHNICAL UNIVERSITY OF MUNICH

SCHOOL OF COMPUTATION, INFORMATION AND
TECHNOLOGY
INFORMATICS

Master's Thesis in Informatics

**DRAFT - Thesis Proposal:
Extension of Pacing Diagram and
Pacing Analysis in p:xe**

Rai Trouvain



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Extension of Pacing Diagram and
Pacing Analysis in p:xe**

**Thesis Proposal: Irgendetwas mit Usability
in PIX:E**

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Disclaimer

I confirm that this Master's thesis is my own work and I have documented all sources and material used.

(Date)

(Rai Trouvain)

Acknowledgements

The acknowledgements

Abstract

English Abstract

Zusammenfassung

German Abstract

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1 Thesis Goal/Research Questions

The main goal of this thesis is to expand on the pacing diagram functionality in p:xe to capture and analyze more types of data that might be interesting in the video game design process.

- RQ1: Which data is interesting?
- RQ2: Which analysis of the data is interesting?

2 Approach

2.1 Original Pacemaker Functionality

After familiarizing myself with the existing codebase, the first step will be to port the original pacemaker functionalities. This includes basic path selection and corresponding intensity/gameplay category diagrams, as well as automatic calculation of some experience parameters (Geheeb, Dyrda, and Geheeb 2024).

2.2 Extend Diagram-Based Analysis

2.2.1 Path Selection

In the long run, the diagram should be extended to include path selection with concurrency and nestedness. While the implementation of this might not end up being part of my thesis, the software architecture and any changes I make should be extendable in that regard.

2.2.2 Lock/Key Dynamic

One big feature will be the analysis of lock/key dynamics and skills requirements. The basis of this will be the incorporation of locks/keys in the existing diagrams and the path selection functionality. The latter requires implementation of a specialized pathfinding algorithm like the one proposed by Aversa, Sardina, and Vassos (2015). The path selection would already offer a feasibility test, i.e. whether the lock/key puzzle is even solvable. Other approaches for useful analyses might be a challenge/entertainment rating (Font et al. 2016) or similarity checks that could help prevent potentially boring repetitions.

2.2.3 LLM-based evaluation

As p:xe already includes some LLM-based features, it could be easily extended to offer LLM-based evaluations along paths. Some ideas for this kind of analysis would be a narrative summary or the extraction of experience parameters from beat descriptions.

2.2.4 Potential Further Extensions

There is a time buffer to account for further promising extensions. This includes the functionality of px components or other artifacts.

2.3 Evaluate

One way to evaluate the extended functionalities would be a user study. The details of how to best conduct it are yet to be determined.

3 Timeline

	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Research										
Codebase										
Path Selection										
Lock/Key Dynamics										
Implementation										
Pacemaker Functionality										
Lock/Key Dynamics										
LLM-based Analysis										
Evaluation										
User Study?										
Writing										
Path Selection*										
Previous Functionality*										
Lock/Key Dynamics*										
LLM-based Analysis*										
Evaluation*										
Background										
Future Work										
Introduction + Conclusion										

Final Submission

Jul 31

* includes methodology and results

Bibliography

Davide Aversa, Sebastian Sardina, and Stavros Vassos. 2015. Path Planning with Inventory-Driven Jump-Point-Search. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, 11(1):2–8.

Jose M. Font, Roberto Izquierdo, Daniel Manrique, and Julian Togelius. 2016. Constrained Level Generation Through Grammar-Based Evolutionary Algorithms. In Giovanni Squillero and Paolo Burelli, editors, *Applications of Evolutionary Computation*, pages 558–573, Cham. Springer International Publishing.

Julian Geheeb, Daniel Dyrda, and Sebastian Geheeb. 2024. PaceMaker: A Practical Tool for Pacing Video Games. In *2024 IEEE Conference on Games (CoG)*, pages 1–8.